COURSE GUIDE & SYLLABUS · GRADES 3–5 · AGES 9–11

# Scratch (Gr 3–5) Advanced: Arcade Engineers

Clones, gravity, lists and your own blocks — engineer real arcade games.

The third course in the Scratch pathway. Learners engineer arcade games with the techniques real game makers use: clones for many enemies and items, velocity and gravity for jumping, lists for words, inventories and passwords, custom blocks with inputs, and game states with menus and levels. Three projects (Starfall Catcher, Jumpy Platformer, Guess the Animal) and a capstone original game. Every lesson has a video, a build-along, Code Lab challenges checked automatically, a worksheet and a teacher lesson plan.

## What learners will be able to do

- Create and delete clones that each run their own behaviour
- Use polling loops for smooth game controls
- Model velocity and gravity with variables to make jumping
- Store, read, search and randomly select items in lists
- Define custom blocks with inputs to remove repetition
- Coordinate menus, levels and game over with broadcasts and state variables
- Playtest, tune and improve games based on feedback
- Design, build and present an original arcade game

## How every lesson works

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| **Watch** | A short animated explainer video with captions (and an optional read-aloud). |
| **Learn** | Learn cards with the real coding blocks, and a step-by-step build-along. |
| **Try it (Code Lab)** | Code in the built-in Scratch-style Code Lab with a live stage. Challenges are checked automatically and give a clear next step; hints unlock one at a time. |
| **Puzzles & quick checks** | Robot puzzles and questions that build understanding, with instant feedback. |
| **Probe** | Debugging and “new situation” tasks to show real understanding. |
| **Projects** | Each module ends with a creative project built in the Code Lab and reviewed by the teacher with a rubric. |
| **Printables** | A worksheet for every lesson, a project brief for every project, and a lesson plan with answer keys for the teacher. |

## Scope and sequence

| # | Lesson | Big idea | Learners build |
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| 1 | **Clones: One Sprite, Many Copies**<br>Use clones to make many copies of a sprite that each run their own script, and delete them when they are done. | Clones | Make it rain stars |
| 2 | **Clones with Personality**<br>Give clones different speeds, sizes and reactions, and detect when a clone touches the player. | Clones + variables and sensing | Catch the falling stars |
| 3 | **Project: Starfall Catcher**<br>Plan, build and test a catching game with clones, smooth controls, a score and an ending. | Project: clones game | Project: Starfall Catcher |
| 4 | **Gravity: What Goes Up…**<br>Use a velocity variable to make a sprite fall faster and faster and land on the floor, and give it smooth sideways movement. | Velocity and gravity | Drop the frog |
| 5 | **Jump!**<br>Make a sprite jump by setting its velocity upward, only when it is standing on the floor. | Jumping with velocity | Make the frog jump |
| 6 | **Project: Jumpy Platformer**<br>Plan, build and test a mini platformer with gravity, jumping, walking, a collectable and a goal. | Project: platformer | Project: Jumpy Platformer |
| 7 | **Lists: A Variable That Holds Many Things**<br>Create a list, add items to it, read an item by its position, and pick a random item. | Lists | The Fortune Owl |
| 8 | **Searching Lists**<br>Check whether a list contains something and use it to make decisions, like a password checker or an inventory. | List search | The Secret Door |
| 9 | **Project: Guess the Animal**<br>Plan, build and test a guessing game that picks a secret from a list and lets the player guess until they get it. | Project: lists game | Project: Guess the Animal |
| 10 | **My Blocks: Make Your Own Blocks**<br>Define a custom block with an input, call it several times with different values, and explain how it removes repeated code. | Custom blocks (procedures) | The square machine |
| 11 | **Game States and Levels**<br>Use broadcasts and variables to add a start screen, levels and a game-over state to a game. | Game states | Menu, levels, game over |
| 12 | **Capstone: Arcade Engineer**<br>Design, build, test and present an original arcade game that uses clones, variables, a custom block or list, and game states. | Capstone project | Project: Arcade Engineer |

## Modules

**Module 1 · Clones**
One sprite, many copies: falling stars, catching games and tidy clean-up. Project: Starfall Catcher.

Big question: How do games make so many things at once?

**Module 2 · Game physics**
Velocity, gravity and jumping with smooth controls. Project: Jumpy Platformer.

Big question: How do games make movement feel real?

**Module 3 · Lists**
Store, read, search and randomly pick items. Project: Guess the Animal.

Big question: How can a program remember lots of things?

**Module 4 · Engineer it**
Custom blocks, game states and levels, then a capstone original arcade game.

Big question: How do engineers keep big programs organised?

## Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and the four projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

## What you need

- Laptop or Chromebook with a modern browser (a keyboard helps for games)
- The built-in Kitabash Code Lab (no account needed)
- Optional: Scratch (scratch.mit.edu or the Scratch app)

## Before you start

- Scratch (Gr 3–5) Intermediate: Story, Quiz & Music Studio, or confident use of loops, variables, broadcasts and ask/answer

## Safety and privacy

Games never ask for real names, addresses or passwords — password lessons use made-up words only. The Code Lab runs inside Kitabash; sharing to scratch.mit.edu is optional and needs a parent-approved account for under-13s. Learners under 18 use the AI tutor only with guardian consent.

## What comes next

After this course: **Scratch (Gr 6–8) Beginner**, which goes deeper into clones, custom blocks, lists and bigger games.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.